Experiment 4

Virtual Labs

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Minimum Spanning Trees

Estimated Time

15 minutes

Learning Objectives of this Module :

In this module, we will :

- Gain the intuition for Kruskal's Algorithm
- Learn when and how to add an edge to MST
- Learn about the primitive Kruskal's algorithm
- Practice the algorithm
- Test your conceptual understanding with a short quiz

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Minimum Spanning Trees

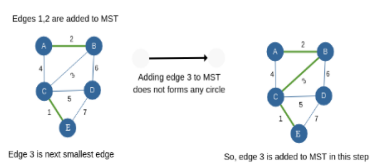
How can we find MST using Kruskal's?

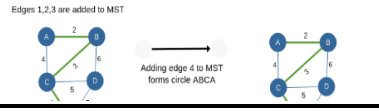
Kruskal's Algorithm is a classic method for finding the Minimum Spanning Tree (MST) of a graph.

- First, sort all edges of the graph in **increasing** order of their weights.
- Check each edge in sorted order to see if it should be included in the MST.
- If adding an edge would create a cycle (detected using Union-Find), skip it; otherwise, add it to the MST.
- Repeat until the MST contains exactly N-1 edges (where N is the number of vertices).

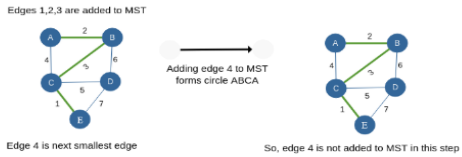
> **Note:** Each edge is considered only once, and the MST is built by always choosing the smallest available edge that does not form a cycle.

When should we add an edge to the MST?





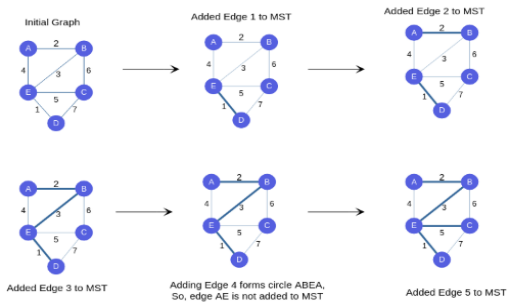
Feedback



Important Observations

- After each iteration, the smallest available edge that does not form a cycle is added to the MST.
- After N-1 edges are included, the MST is complete.
- Work through each step and see which edges are included.

Step by Step Process for Kruskal's



> **Tip:** Make sure all vertex names in your diagrams are unique to avoid confusion.

Concept of Kruskal's Algorithm

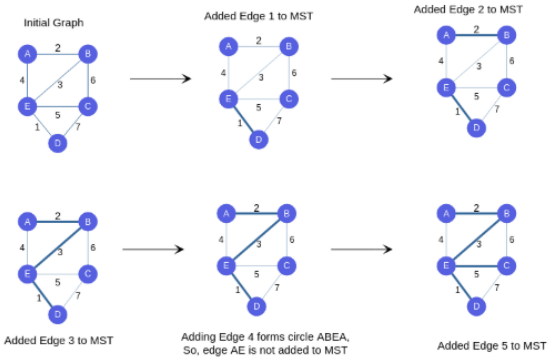
Let's have a final look at the consolidated algorithm to find MST of given graph:

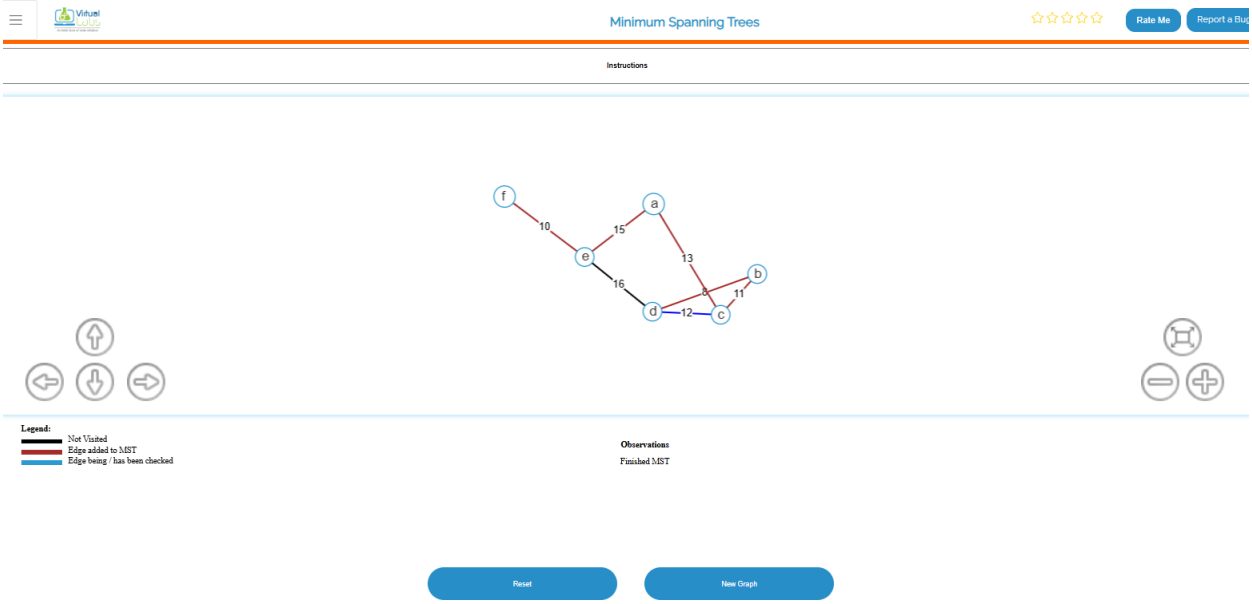
- **STEP 1 :** Sort the given edges
- **STEP 2 :** Check each edge in sorted order if it forms a cycle with already selected edges. If not Add it to list of MST . If it does move to next edge.
- **STEP 3 :** Run steps 1 and 2 till v-1 edges are selected (v=no.of.vertices).

Observations

- From the mentioned Algo, we can conclude that after the Tth iteration, we will have the edges which should be in MST among T smallest places included in MST.
- So, After N iterations we will have all edges which are to be in MST included in it.
- Notice that after including N-1 edges in MST , MST will be finished.
- Look at the picture below and work out the result of each iteration. See if it matches the picture, and notice which elements keep getting placed correctly after each iteration!

Iteration by Iteration Visualization of Kruskal's

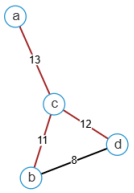



Minimum Spanning Trees
Rate Me
Report a Bug
Instructions
f
a
e
b
d
c
10
15
13
16
8
11
12
Legend:
Not Visited
Edge added to MST
Edge being / has been checked
Observations
Finished MST
Reset
New Graph

Minimum Spanning Trees
Rate Me
Report a Bug
Instructions
b
a
c
d
e
8
10
11
12
Legend:
Not Visited
Edge added to MST
Edge discarded
Edge being checked
Min.Speed
Max.Speed
Observations
Finished adding n-1 edges MST is formed by the edges in brown
Next
New Graph
Reset
Pause

Minimum Spanning Trees

Rate Me



Debug Info
Selected edges: [13, 12, 11]

Observations

Selected spanning tree is correct.

Reset | New Graph | Submit

Minimum Spanning Trees

Choose difficulty: ☑ Beginner ☑ Intermediate ☑ Advanced

1. What is incorrect about Kruskal's algorithm?
- a: Edges in MST cannot form cycles Explanation
- b: Kruskal's algorithm uses greedy approach Explanation
- c: It makes use of disjoint set data structure Explanation
- d: MST is constructed from decreasing weighted edges Explanation

2. In a graph having V vertices and E edges, what is the smallest possible edge count of a Minimum Spanning Tree?
- a: E-1 Explanation
- b: V+1 Explanation
- c: V-1 Explanation
- d: E Explanation

3. How do we detect whether one edge(with V1 and V2 as vertices on both the ends) is a part of a cycle in the graph?
- a: V1 and V2 belong to different set in union-find Explanation
- b: V1 has smaller value than V2 Explanation
- c: V1 and V2 belong to same set in union-find Explanation
- d: V1 has larger value than V2 Explanation

4. Which edge gets judged if it should or shouldn't be in MST after n iterations?
- a: Smallest edge Explanation
- b: Largest Edge Explanation
- c: Middlemost Edge in increasing order Explanation
- d: None of the above Explanation

5. When do we add an edge to MST while running Kruskal's ?
- a: When edge forms a cycle in MST Explanation
- b: When edge doesn't forms a cycle in MST Explanation
- c: When edge is the next smallest possible edge irrespective of the cycle Explanation
- d: None of the above Explanation

Submit Quiz

Score: 5 out of 5

Conclusion:

For Faculty Use

Correction Parameters	Formative Assessment [40%]	Timely completion of Practical [40%]	Attendance / Learning Attitude [20%]	
Marks Obtained				